You have two inputs:

1) Clear or reset (it's btnC on basys3 board). When this input is 1 (or the button pressed) it resets the current run.
2) A switch (sw[0] on basys3 board), input variable x.

You have a 16 bit output, corresponding to 16 LEDs on basys3 board (LED [0:15]).

- When x == 0, output will follow the pattern below continuously (within 1s intervals), <after 16th state continues from the first>

0000000000000001

0000000000000010

0000000000000100

0000000000001000

0000000000010000

0000000000100000

0000000001000000

0000000010000000

0000000100000000

0000001000000000

0000010000000000

0000100000000000

0001000000000000

0010000000000000

0100000000000000

1000000000000000

- When x == 1, output will follow the pattern above in reverse order continuously (within 1s intervals), <after 16th state continues from the first>

NOTE: Don't write each number as above, itS not smart.

Task1: compose a design source

Task2: write a test bench file

Task3: modify your design source description for Basys3.

Use timer sub-module:

```
//////////////////////////////////////////////////////////////////////////////////
module zaman(
output Y,
input clock,
input reset
);
reg elapsed;
reg [25:0] state;
always @(posedge clock)
if (reset == 1) state <= 0;
else if (state == 50000000) state <= 0;// corresponds to 1 sec
else state <= state + 1;
always @(state)
if (state == 50000000) elapsed = 1;
else elapsed = 0;
assign Y = elapsed;
endmodule
//////////////////////////////////////////////////////////////////////////////////
```

YOUR REPORTS:

Include all your codes and simulation results.

When requires explain your descriptions and results.